Republic of the Philippines
NORTHWEST SAMAR STATE UNIVERSITY
Calbayog City, Samar

Junior Philippine Institute of Civil Engineers

Activity Proposal: Inter-Year Bridge Design Competition 2026

Proponent	Junior Philippine Institute of Civil Engineers
Date	October 24, 2026, 8:00 AM to 5:00 PM
Venue	CEA Laboratory Building, NwSSU Main Campus
Participants	60 civil engineering students in 12 teams (first to fourth year)
Academic year	2026-2027, First Semester

I. Rationale

Structural analysis is taught mostly on paper, and students rarely see how a design behaves under load. A design-and-build competition lets teams apply statics and mechanics of materials to a small truss bridge, then test it to failure. The activity also builds teamwork across year levels, since each team mixes junior and senior students.

II. Objectives

- Apply truss analysis and member sizing to a physical model.
- Compare predicted and actual failure loads, and explain the difference.
- Encourage collaboration between year levels.

III. Program of Activities

Time	Activity	In charge
8:00 AM	Registration and briefing on rules and safety	Secretary
8:30 AM	Design and construction (balsa wood, 1.2 m span)	Team leaders
1:00 PM	Load testing with calibrated weights	Engr. Ana Santos
3:30 PM	Presentation of design reports	Teams
4:30 PM	Awarding and closing remarks	President

IV. Budgetary Requirements

Item	Qty	Unit cost	Amount
Balsa wood sticks (bundle)	24	PHP 180.00	PHP 4,320.00
Wood glue (250 ml)	12	PHP 95.00	PHP 1,140.00
Cutting mats and blades	12	PHP 150.00	PHP 1,800.00
Certificates and printing	1	PHP 600.00	PHP 600.00
Prizes for top three teams	3	PHP 1,000.00	PHP 3,000.00
Total			**PHP 10,860.00**

Source of funds: JPICE organizational fund, AY 2026-2027

V. Expected Outcomes

- Twelve tested bridge models with written design reports.
- A record of predicted versus actual failure loads for use in class discussions.

Prepared by:

Juan Dela Cruz
President, JPICE

Noted by:

Engr. Ana Santos
Adviser, JPICE